

GroupTrack

Firebase SHA-256 Fix

Resolves 7–10 Second Startup Delay

May 2, 2026 | Continuation Task

Problem

GroupTrack experiences a 7–10 second delay on app startup, showing the Meshtastic logo screen and occasionally a "Getting Started" onboarding screen. The app is unresponsive during this delay. This affects both release and debug builds on all devices.

Root Cause

Diagnosed via adb logcat. Google Play Services is rejecting the app's signing certificate:

```
SecurityException: GoogleCertificatesRsult: not allowed: pkg=com.grouptrack.android, sha256=35b05fe0c752f0a1de8ffbf793e31ad9ec703d621280b6067918c2f5b7873dc5
```

The google-services.json was updated for com.grouptrack.android in commit 744c1588c (Play Store prep), but the SHA-256 fingerprints for both the debug and release signing keys were never registered in Firebase. Google Play Services tries to authenticate on every app launch, gets rejected, and waits for timeout — that's the 7–10 second delay.

SHA-256 Fingerprints to Register

Key	SHA-256 Fingerprint
Release	37:F0:13:67:D4:54:FD:F0:47:66:6B:A6:D7:6F:5D:E7:26:51:7C:7C:73:82:0F:23:52:D9:8C:51:C7:AB:95:00
Debug	35:B0:5F:E0:C7:52:F0:A1:DE:8F:FB:F7:93:E3:1A:D9:EC:70:3D:62:12:80:B6:06:79:18:C2:F5:B7:87:3D:C5

Release fingerprint obtained from: `keytool -list -v -keystore C:/Users/kixaz/grouptrack-release-key.jks -alias grouptrack`

Debug fingerprint obtained from: logcat SecurityException error message

Solution — Step by Step

Part A: Add Fingerprints in Firebase Console

Open: <https://console.firebase.google.com>

1. Sign in with your Google account
2. Select the GroupTrack project (or the Meshtastic project that contains com.grouptrack.android)
3. In the left sidebar, click the gear icon (⚙️) next to "Project Overview"
4. Click "Project settings"
5. Scroll down to the "Your apps" section

- Find the Android app entry for com.grouptrack.android — if it doesn't exist, you may see the original Meshtastic package. In that case, click "Add app" and add com.grouptrack.android as a new Android app.
- Under the app entry, find "SHA certificate fingerprints"
- Click "Add fingerprint"
- Paste the RELEASE fingerprint (copy exactly, including colons):

```
37:F0:13:67:D4:54:FD:F0:47:66:6B:A6:D7:6F:5D:E7:26:51:7C:7C:73:82:0F:23:52:D9:8C:51:C7:AB:95:00
```

- Click Save
- Click "Add fingerprint" again
- Paste the DEBUG fingerprint:

```
35:B0:5F:E0:C7:52:F0:A1:DE:8F:FB:F7:93:E3:1A:D9:EC:70:3D:62:12:80:B6:06:79:18:C2:F5:B7:87:3D:C5
```

- Click Save

Part B: Download Updated google-services.json

- Still on the Project Settings page, scroll to the app entry
- Click the "google-services.json" download button
- Save the file to Downloads

Part C: Replace google-services.json in Project

In local Git Bash:

- Copy new file into project:

```
cp ~/Downloads/google-services.json ~/Meshtastic-Android/app/google-services.json
```

- Verify it's updated:

```
grep -c "grouptrack" ~/Meshtastic-Android/app/google-services.json
```

Should return a number > 0, confirming the GroupTrack package is in the file.

Part D: Rebuild and Test

Release build (Android 1):

```
./gradlew assembleGoogleRelease -x uploadCrashlyticsMappingFileGoogleRelease
adb -s 8624SBCEDF00001789 install -r -d app/build/outputs/apk/google/release/app-google-release.apk
adb -s 8624SBCEDF00001789 shell am force-stop com.grouptrack.android
```

Open the app — should load to convoy map in under 3 seconds with no Meshtastic logo delay.

If the fix works, commit and build AAB:

```
git add app/google-services.json
git commit -m "fix: updated google-services.json with release+debug SHA-256 fingerprints"
./gradlew bundleGoogleRelease -x uploadCrashlyticsMappingFileGoogleRelease
```

Upload new AAB to Play Console → Internal Testing → Create new release.

Verification

After the fix, confirm the `SecurityException` is gone:

```
adb -s 8624SBCEDF00001789 logcat -c
```

Open the app, then:

```
adb -s 8624SBCEDF00001789 logcat -d | grep -i "SecurityException.*grouptrack"
```

Should return no results. If it still appears, the fingerprint was entered incorrectly — double-check for copy/paste errors in the colons.

Key Information

Item	Value
Package	com.grouptrack.android
Firebase console	https://console.firebase.google.com
Keystore location	C:/Users/kixaz/grouptrack-release-key.jks
Keystore alias	grouptrack
Keystore password	Sports256!
google-services.json	~/Meshtastic-Android/app/google-services.json
Commit that caused this	744c1588c — updated google-services.json for com.grouptrack.android without adding SHA-256 fingerprints
Android 1 serial	8624SBCEDF00001789
Android 2 serial	24039703201775

Also Completed May 1–2 Session

- Mileage display fix — shipped and working (ConvoyGpsService.kt)
- On-demand trail loading — 22MB GeoJSON reads on TRAILS tap only (both maps)
- WebView cleanup — DisposableEffect destroys WebView on MapViewer exit
- Play Store AAB v29320458 published to internal testing
- Beta signup form live on grouptrack.org with MySQL backend
- CSV export for Play Console: grouptrack.org/beta_signup.php?export=csv&key=gt_beta_export_2026
- Beta tester flyer created (PDF + PNG for Facebook)
- User manual and install guide PDFs updated on website
- Download links fixed on grouptrack.org
- Android 2 updated with V3.0 build
- V2.5 Release Planning v3 document completed
- Single-node tick guard attempted and reverted — V2.5 task